

Total No. of Questions : 12]

SEAT No. :

P1517

[5460]-108

[Total No. of Pages : 3

T.E. (Civil)

**FOUNDATION ENGINEERING
(2012 Pattern) (301009) (End Sem.)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10 and Q.11 or Q.12.
- 2) Neat diagrams must be drawn whenever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary and mention it clearly.
- 5) Use of non-programmable calculator is allowed.

SECTION - I

Q1) Discuss Seismic refraction method of soil exploration in accordance with:[7]

- a) Principle
- b) Procedure and sketch
- c) Limitation

OR

Q2) a) Differentiate between SPT and DCPT. [3]

b) Discuss factors affecting sample disturbance. [4]

Q3) a) Explain the modifications suggested by Mayerhoff in the Terzaghi's bearing capacity equation. [3]

b) Discuss how to determine bearing capacity of the layered soil. [3]

OR

Q4) A strip footing 1m wide, with the base located at the depth of 1m below ground surface. Soil properties are $\gamma = 18.5 \text{ kN/m}^3$, $c = 3.0 \text{ kN/m}^2$ and $\phi = 20^\circ$. Determine safe bearing capacity using F.S. = 3. Assume soil fails by local shear, for $\phi = 20^\circ$, $N_c = 11.8$, $N_q = 3.9$ and $N_\gamma = 1.7$. Use Terzaghi's analysis. [6]

P.T.O.

- Q5)** Explain the terms: [7]
- a) Compression index.
 - b) Over consolidation ratio
 - c) Allowable settlement
 - d) Consolidation settlement

OR

- Q6)** The consolidation test is conducted on the soil with following properties, compression index 0.25, void ratio at the stress of 10 kN/m² is 2.02, and coefficient of permeability is 3.4×10^{-7} mm/sec. Determine. [7]
- a) Change in void ratio if stress is increased by 9 kN/m².
 - b) Settlement if soil is 4m thick.
 - c) Time required for 40% consolidation for one way drainage. (T_v) = 0.125664.

SECTION - II

- Q7)** a) Enlist the methods of determining pile capacity. Explain any two methods. [6]
b) Sketch a sectional elevation of well foundation. Explain the function of each part. [6]
c) Explain in detail negative skin friction on piles. [4]

OR

- Q8)** a) Find out group capacity of piles by any of following three methods. [8]
i) Fled's rule
ii) Block Failure
iii) Converse Labbare's Formula
iv) Individual
Pile group consisting of 15 piles arranged in 3 rows, diameter of pile is 300 mm, depth of pile 8m, $c = 25\text{kN/m}^2$, Spacing of pile = 0.8 m c/c. $\alpha = 1.0$, Unit weight of soil = 10 KN/m³.
b) Explain with figures the following difficulties and their rectification. [8]
i) Tilt
ii) Hanging up
iii) Sand blow

- Q9)** a) Explain the terms with sketches. [6]
i) Free earth support
ii) Fixed earth support in connection with anchored sheet piles for their bending moment.
b) Explain merits and demerits of Circular type, Diaphragm type cellular cofferdams. [6]
c) What are the various ground improvement techniques? Explain any one of them. [4]

OR

- Q10) a)** Discuss the factors affecting selection of type of Cofferdam and explain the suitability of various types of cofferdams. [6]
- b) What is swelling potential of soil and what are the methods for determination of swelling potential of black cotton soil. [6]
- c) A sheet pile has following details: [4]
- Embedment depth –5m
 - Cantilever height 4m
 - $\phi = 30^\circ$ determine factor of safety against overturning.
- Q11) a)** What is liquefaction? Explain the liquefaction susceptibility criteria. [6]
- b) Explain the use of geosynthetics in
- Retaining wall
 - Deep foundation
 - Embankments on soft soils. [6]
- c) What is reinforced earth wall? Explain with neat sketch. [6]
- OR
- Q12) a)** Define the following terminologies correlated with earthquake. [6]
- Epicenter
 - Focus
 - Focal depth
 - Epicentral distance
 - Foreshocks and aftershocks
 - Body waves
- b) What is geosynthetics? Enlist the advantages of geosynthetics materials for reinforcement and What are its functional requirements? [6]
- c) Enlist and explain different types of seismic waves. [6]
